# Business Intelligence Systems

## INFO 4555 / 5556

Fall 2025

Procurement




Cornell University

# Procurement

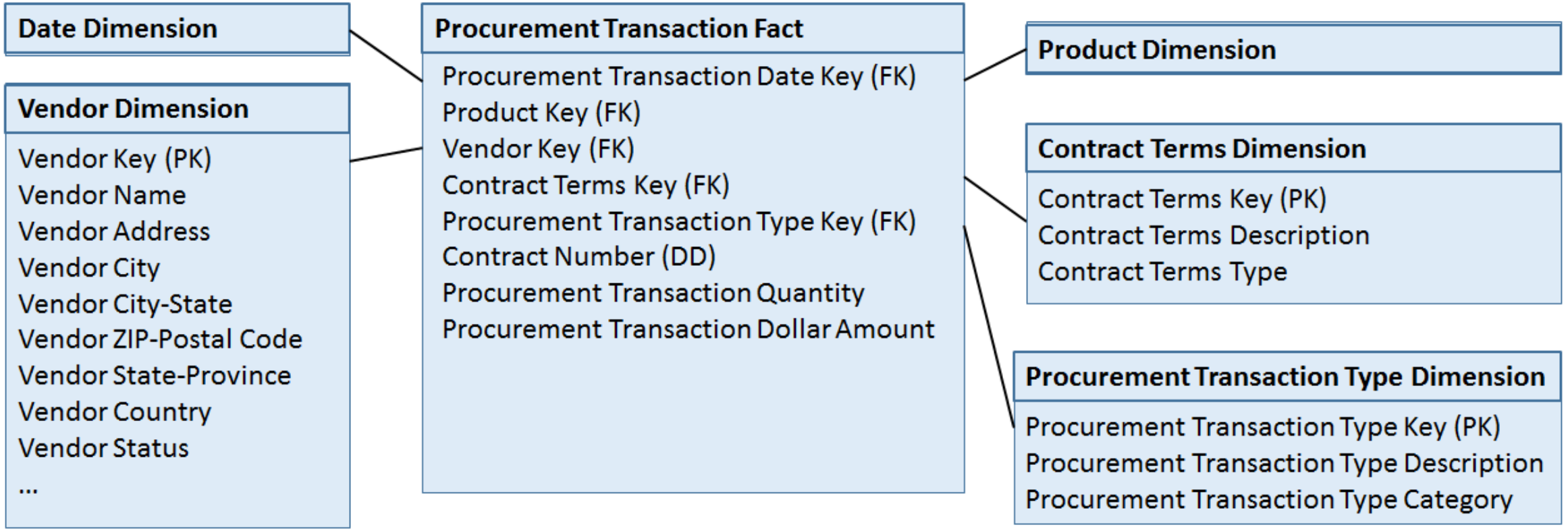
- Procurement is the process of purchasing goods or services in support of the organization
- Procurement is often a critical business function
  - Purchasing raw materials for manufacturing
  - Purchasing items for resale
  - Purchasing equipment in support of the business
  - Purchasing services in support of the business
- It is important to get the appropriate materials and services in the most economical manner.



# Sample analytic requirements for Procurement:

- Which materials or products are most frequently purchased?
  - At what prices?
  - View demand across entire organization, not a single location
- Are employees purchasing from the preferred vendors?
- Are you receiving the negotiated price from your vendors, or is there a variance?
- How are your vendors performing?
  - What is the vendors fill rate?
  - On-time delivery performance?
  - Late deliveries outstanding?
  - Percent back ordered?
  - Reject rate based on receipt inspection?

# Kimball Procurement Transaction Model Example:




*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-1*

# A few Business Processes within Procurement

- Purchase Requisitions
- Purchase Orders
- Shipping Notifications
- Warehouse Receipts
- Vendor Invoices
- Vendor Payments

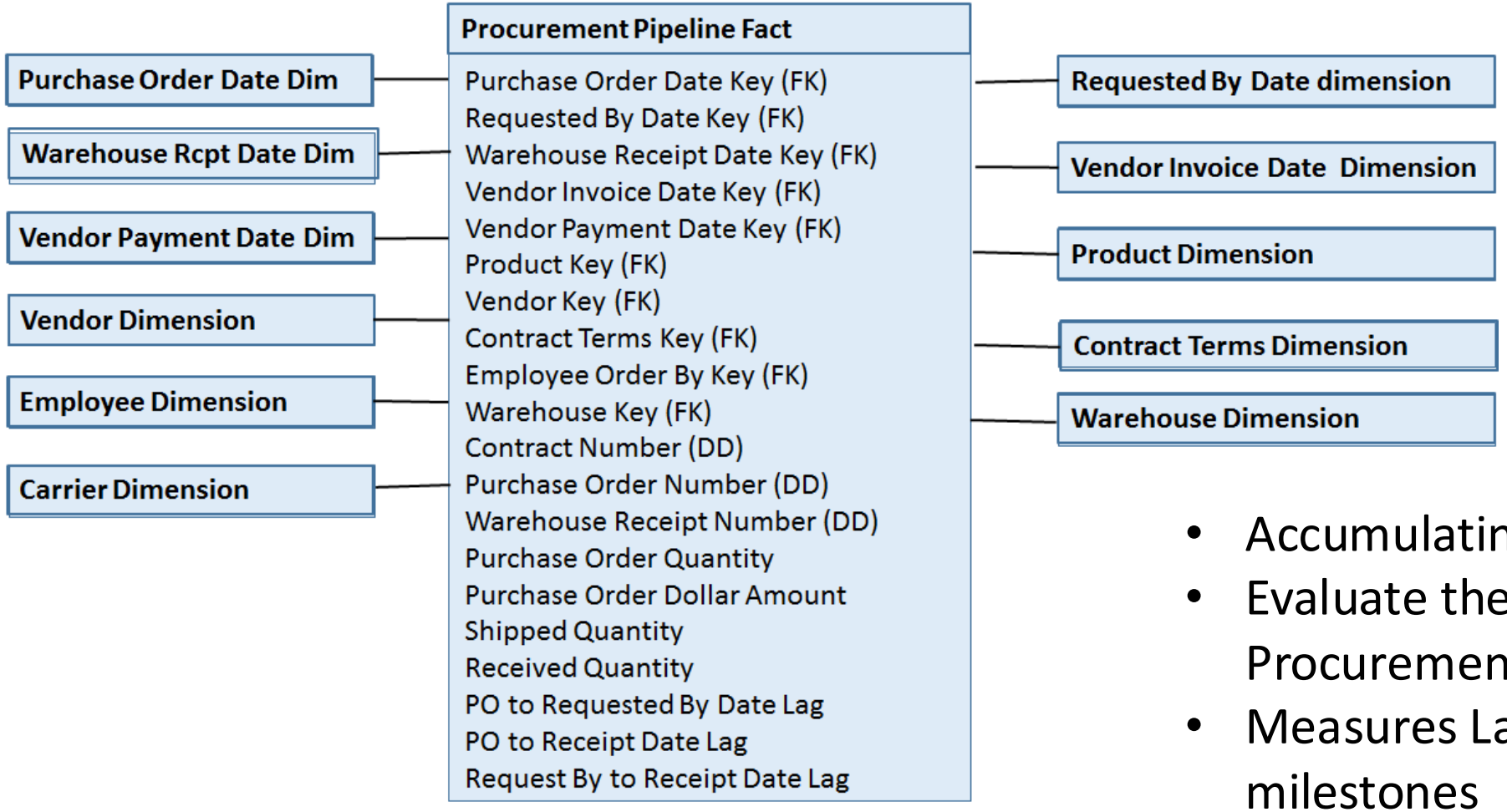
# Single Versus Multiple Transaction Fact Tables

- May only combine transaction types that have the same grain.
- The dimensionality may be different for the transaction types
  - Ex. Discounts taken may apply to vendor payments but no where else
  - Ex. Employee who received shipment may not apply to other transaction types
  - Ex. Each step in the pipeline might have differing natural keys associated with the process
- Need to decide whether to model these transaction types in a single fact or break into multiple fact tables to preserve their unique dimensionality.

# Should Transaction Fact be broken into multiple Fact tables?

- Your modeling decisions should be based on an understanding of the business  requirements AND the realities of what the source data will support.
- Considerations to help make decision to whether you split the facts across multiple tables:
  - What are the users' analytic requirements?
  - Are there really multiple unique business processes?
  - Are multiple source systems capturing metrics with unique granularities?
  - What is the dimensionality of the facts?

# Kimball Procurement Accumulating Snapshot




*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-4*

- Accumulating snapshot
- Evaluate the Performance of the Procurement Pipeline
- Measures Lag Time between milestones

# Kimball Bus Matrix rows for Procurement

| Business Processes | Atomic Granularity | Metrics | Date | Product | Vendor | Control Terms | Employee | Warehouse | Carrier |
|---|---|---|---|---|---|---|---|---|---|
| Purchase Requisitions | 1 row per requisition line | Requisition Quantity & Dollars | X | X | X | X | X | | |
| Purchase Orders | 1 row per PO line | PO Quantity & Dollars | X | X | X | X | X | X | X |
| Shipment Notifications | 1 row per shipping notice line | Shipping Quantity | X | X | X | | X | X | X |
| Warehouse Receipts | 1 row receipt line | Received Quantity | X | X | X | | X | X | X |
| Vendor Invoices | 1 row per invoice line | Invoice Quantity & Dollars | X | X | X | X | | X | |
| Vendor Payments | 1 row per payment | Invoice, Discount & Net Payment Dollars | X | X | X | X | | X | |

- Added two columns to Bus Matrix
  - Atomic Granularity
  - Metrics
- Shows differing Dimensionality between processes
- Shows differing Grains between processes

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-2*

When poll is active, respond at **pollev.com/info4555**
Text **INFO4555** to **22333** once to join

# All of the following are reasons you should consider splitting multiple transaction types into multiple fact tables, except:

A: Transaction types are at differing grains

B: Transaction types have differing names

C: Transaction types have differing dimensionality

D: Transaction types have differing measures

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

# Slowly Changing Dimensions

- In general Dimension Attributes should not change frequently
- Many Dimension Attributes will change over time, so you need to know how they should be handled
- The Kimball Methodology offers 8 options for handling Dimension Attribute Changes.
- Known as Slowly Changing Dimensions (SCD) Types 0 – 7
- decision of how to handle these changes should be driven by business need and include data governance

# SCD – Type 0: Never Changes

- Value Never changes
- Good if you want to maintain the “original value”
- Applies to most attributes in a date dimension
- Date key would be type0 – ex. 20160912, should always be relevant
- Also, the “durable supernatural keys” introduced in chapter 3 would be type 0 – guaranteed to never change

# SCD – Type 1: Overwrite

- The old value is simply overwritten by the new value
- Value always represents the most recent assignment
- Easy to implement, works well with “full table refresh” model
- Lose ability to track change
- any aggregations build off a TYPE 1 dimension should be rebuilt

# SCD – Type 1: Overwrite (example)

Original row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name |
|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education |

Updated row in Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name |
|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Strategy |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-5*



# SCD – Type 2: Add new row

- Primary method for preserving dimension attribute history in a data warehouse
- Three new attributes added to the dimension table: effective_date, expiration_date, current_value_indicator
- When a value changes, a new row is entered into the dimension table.
  - A new surrogate (PK) is assigned, the current_value_indicator is set to current
  - The prior record's expiration_date is set to a period prior to the new record
  - The prior record's current_record_indicator field is set to "expired"
- The fact table will join to the appropriate values for that particular time, due to the differing surrogate key values

# SCD-Type 2: Add new row (Example)

Original row in the Product dimension:

| Product Key | SKU (NK) | Product Description | Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|---|---|---|---|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education | ... | 2012-01-01 | 9999-12-31 | Current |

Rows in Product dimension following department reassignment:

| Product Key | SKU (NK) | Product Description | Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|---|---|---|---|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education | ... | 2012-01-01 | 2013-01-31 | Expired |
| 25984 | ABC922-Z | IntelliKidz | Strategy | ... | 2013-02-01 | 9999-12-31 | Current |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-6*

**Event: Department Name Changed**

Steps Taken by ETL

- New record inserted into Dimension table
- New surrogate key assigned
- Prior record "expiration date" set
- Prior record "current row indicator" set to "expired"
- New record "expiration date" set to date far in the future

When poll is active, respond at **pollev.com/info4555**
Text **INFO4555** to **22333** once to join

# How are you feeling about Slowing Changing Dimensions so far?

A: Great – Easy, keep on going with the lecture

B: OK - I'm following you, keep going

C: Bad – What are you talking about, need a quick review of types 0-2

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

# SCD – Type 3: Add new attribute

- When there is a need to see the fact data as if the dimension value change had never occurred
- The primary (original) column is treated as a type 1 (changed to new value) the prior value is added to a column designated for the prior value.  Such as “prior department name”
- Allows existing reports to use current values without any change
- may query the “prior value column” to see the historical view
- most appropriate for changes that impact a large portion of the dimension records and users want to assess the impact of the change
- Not good for attributes which change frequently. Only allows the current and prior versions.



# SCD – Type3: Add new attribute (Example)

Original row in Production dimension:

| Product Key | SKU (NK) | Product Description | Department Name |
|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education |

Row row in Production dimension:

| Product Key | SKU (NK) | Product Description | Department Name | Prior Department Name |
|---|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Strategy | Education |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-8*

**Event: Department Name Change**

Steps taken by ETL:

- Prior department name added to "prior_department_name" attribute
- "product description" attribute set to new department name

# SCD – Type4: Add mini dimension

- break out the frequently changing attributes into their own “mini dimension”
- For values which change frequently, a type 2 dimension could get overwhelmingly large
- mini dimension would likely contain all of the possible combinations for the attributes (often use ranges)
- if the customer’s demographic profile changes, the next transaction added to the fact would join to the new appropriate mini-fact dimension.

# SCD – Type 4: Add mini dimension (Example)

| Demographics Key | Age Band | Purchase Frequency Score | Income Level |
|---|---|---|---|
| 1 | 21-25 | Low | <$30,000 |
| 2 | 21-25 | Medium | <$30,000 |
| 3 | 21-25 | High | <$30,000 |
| 4 | 21-25 | Low | $30,000-39,000 |
| 5 | 21-25 | Medium | $30,000-39,000 |
| 6 | 21-25 | High | $30,000-39,000 |
| … | … | … | … |
| 142 | 26-30 | Low | <$30,000 |
| 143 | 26-30 | Medium | <$30,000 |
| 144 | 26-30 | High | <$30,000 |
| … | … | … | … |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-10*

- Mini dimension contain a record for every possible combination of the demographic information
- If any of the tracked demographic attributes changes for a customer, the next FACT table record would have the appropriate "demographics key" representing the new valid demographic values

When poll is active, respond at **pollev.com/info4555**
Text **INFO4555** to **22333** once to join

# According to Kimball, what is the most common SCD type used to track Dimensional attribute change history?

A: Type 0

B: Type 2

C: Type 4

D: Type 7

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

# SCD – Types 5-7 are Hybrids

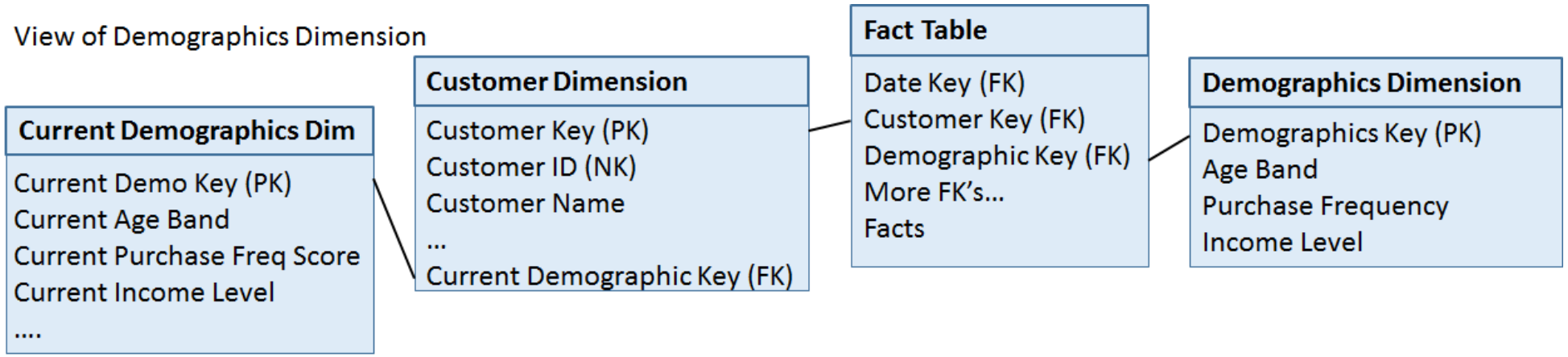
- add greater analytic flexibility
- Often create more complexity for usage of the resulting models
- make sure the users are willing to accept the additional complexity
- Remember a primary goal of dimensional modeling is the ease of use for the business users

# SCD – Type 5: Mini-Dimension and Type 1 Outrigger

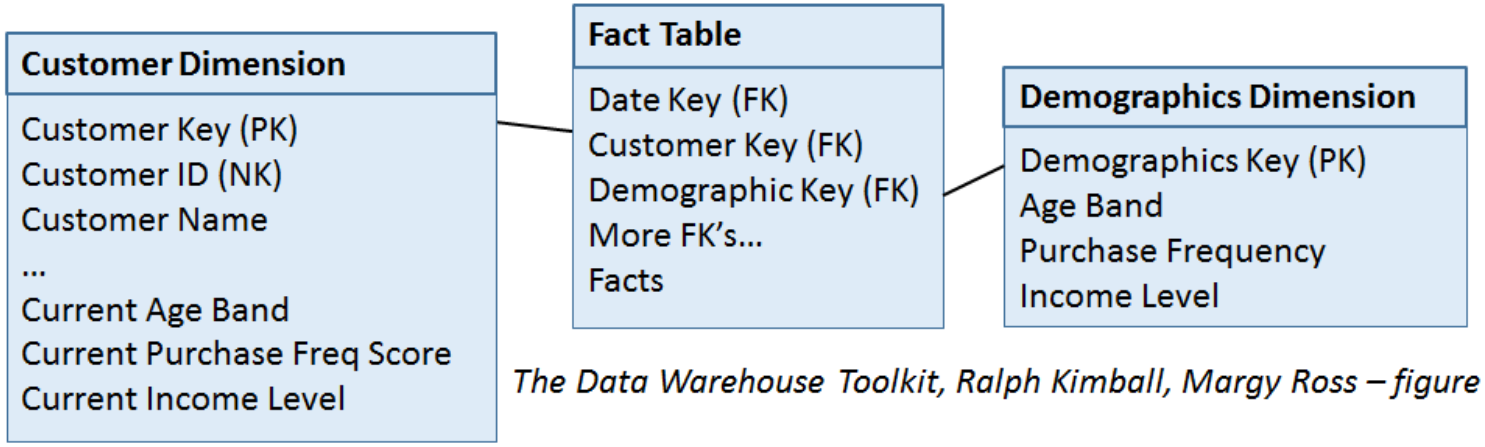
- mini-dimension table is joined to a primary dimension as an outrigger.
- The join is handled as a type 1, so only represents the current value
- allows the association of the mini-dimension values to the primary dimension, without the need to go through a fact table.
- Allows the extension of attributes for the primary dimension without storing many repeating and changing values
- mini-dimension tied to the fact table gives you the historic reference

# SCD – Type 5: (Example)

View of Demographics Dimension




Logical representation to the BI tools:




*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-12*

# SCD – Type 6: Add Type 1 Attributes to Type 2 Dimension

- combination of Type 2 and Type 3
- creates new records, with new surrogate keys
- also contains a “current attribute” fields, so historical queries may also see current value

# SCD – Type 6: (Example)

Original row in Product dimension:

| Product Key | SKU (NK) | Product Description | Historic Department Name | Current Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|---|---|---|---|---|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education | Education | ... | 2012-01-01 | 9999-12-31 | Current |

Rows in in Product dimension following first department reassignment:

| Product Key | SKU (NK) | Product Description | Historic Department Name | Current Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|---|---|---|---|---|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education | Strategy | ... | 2012-01-01 | 2013-01-31 | Expired |
| 25984 | ABC922-Z | IntelliKidz | Strategy | Strategy | ... | 2013-02-01 | 9999-12-31 | Current |

Rows in in Product dimension following second department reassignment:

| Product Key | SKU (NK) | Product Description | Historic Department Name | Current Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|---|---|---|---|---|---|---|---|---|
| 12345 | ABC922-Z | IntelliKidz | Education | Critical Thinking | ... | 2012-01-01 | 2013-01-31 | Expired |
| 25984 | ABC922-Z | IntelliKidz | Strategy | Critical Thinking | ... | 2013-02-01 | 2013-06-30 | Expired |
| 31726 | ABC922-Z | IntelliKidz | Critical Thinking | Critical Thinking | ... | 2013-07-01 | 9999-12-31 | Current |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-13*

**Example of two department reassignments:**

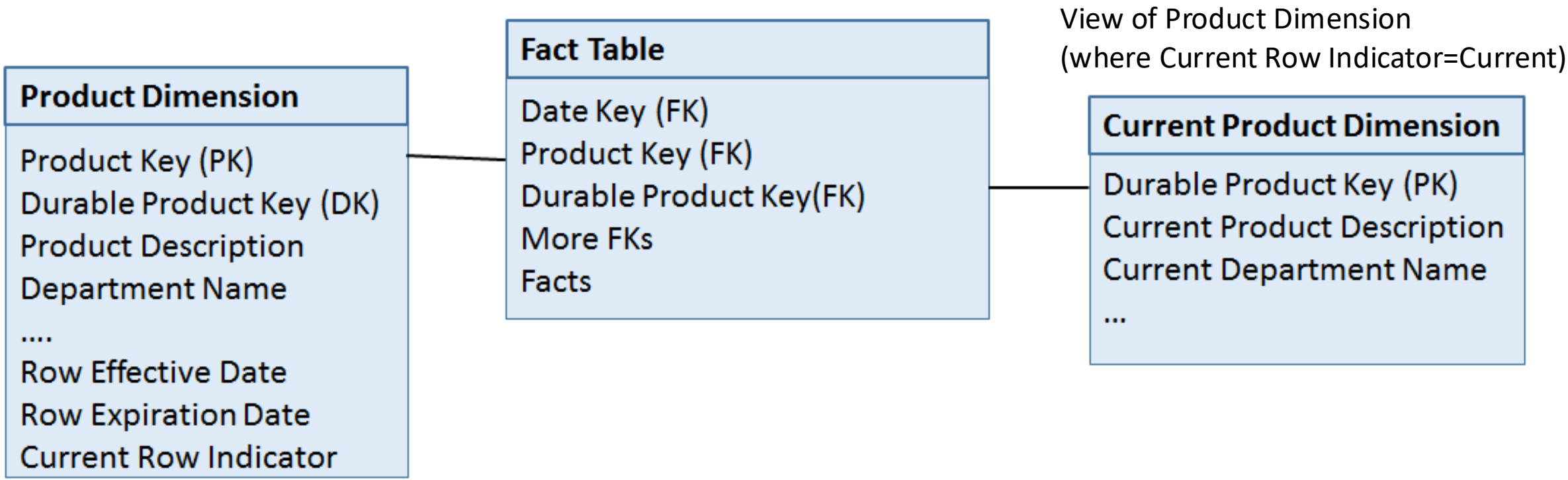
ETL steps:

- Add new record to the product dimension table (new surrogate key assigned)
- The "current department name" attribute set to current name for ALL records related to this SKU (NK)
- Prior record "current row indicator" set to "expired"
- Prior record "row expiration date" set to date immediately before new record "effective date"

# SCD – Type 7: Dual type 1 and 2 Dimensions

- base dimension contains a type 2 primary key and other type 2 supporting attributes such as effective_date, expiration_date, current_row_indicator
- base dimension contains a durable key (DK) in addition to the Primary key. The durable key does not change with the type 2, it will remain the same for the life of the warehouse
- fact table contains foreign keys for both the TYPE 7 primary key (type 2) and the durable key (DK)
- queries using the base dimensions would handle like a typical Type 2 dimension and represent values associated with the appropriate point in time for the joined fact table
- Kimball suggests creating a role-play view of the base dimension, with the current indicator set to current and joining to the fact table via the durable key. This will result in a current values only view of the data associated with the fact table
- Type 7 SCD can support “As Of” Reporting. Reporting on various past rollups and comparing them to current value rollups

# SCD – Type 7: (Example model)




*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-14*

# SCD – Type 7: (Example Data)

Rows in Product dimension:

| Product Key | SKU (NK) | Durable Product Key | Product Description | Department Name | ... | Row Effective Date | Row Expiration Date | Current Row Indicator |
|---|---|---|---|---|---|---|---|---|
| 12345 | ABC922-Z | 12345 | IntelliKidz | Education | ... | 2012-01-01 | 2013-01-31 | Expired |
| 25984 | ABC922-Z | 12345 | IntelliKidz | Strategy | ... | 2013-02-01 | 2013-06-30 | Expired |
| 31726 | ABC922-Z | 12345 | IntelliKidz | Critical Thinking | ... | 2013-07-01 | 9999-12-31 | Current |

Rows in Product dimension's current view:

| Product Key | SKU (NK) | Durable Product Key | Current Product Description | Current Department Name | ... |
|---|---|---|---|---|---|
| 12345 | ABC922-Z | 12345 | IntelliKidz | Critical Thinking | ... |
| 25984 | ABC922-Z | 12345 | IntelliKidz | Critical Thinking | ... |
| 31726 | ABC922-Z | 12345 | IntelliKidz | Critical Thinking | ... |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-16*

Note that this is an error in the textbook. The view filters on current record indicator, so there should only be one record for IntelliKidz and the Durable Product key would be unique.

# Slowly Changing Dimension Types

| SCD TYPE | Dimension Table Action | Impact on Fact Analysis |
|---|---|---|
| Type 0 | No change to attribute value. | Facts associated with attribute's original value. |
| Type 1 | Overwrite attribute value. | Facts associated with attribute's current value. |
| Type 2 | Add new dimension row for profile with new attribute value. | Facts associated with attribute value in effect when fact occurred. |
| Type 3 | Add new column to preserve attribute's current and prior values. | Facts associated with both current and prior attribute alternative values. |
| Type 4 | Add mini-dimension table containing rapidly changing attributes. | Facts associated with rapidly changing attributes inn effect when fact occurred. |
| Type 5 | Add type 4 mini-dimension, along with overwritten type 1 mini-dimension key in base dimension. | Facts associated with rapidly changing attributes in effect when fact occurred, plus current rapidly changing attribute values. |
| Type 6 | Add type 1 overwritten attributes to type 2 dimension row, and overwrite all prior dimension rows. | Facts associated with attribute value in effect when fact occurred, plus current values. |
| Type 7 | Add type 2 dimension row with new attribute value, plus view limited to current rows and/or attribute values. | Facts associated with attribute value in effect when fact occurred, plus current values. |

*The Data Warehouse Toolkit, Ralph Kimball, Margy Ross – figure 5-17*



When poll is active, respond at **pollev.com/info4555**
Text **INFO4555** to **22333** once to join

# Which of the Kimball SCD types only maintains current dimension values?

A: Type 0

B: Type 1

C: Type 2

D: Type 3

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

When poll is active, respond at **pollev.com/info4555**

Text **INFO4555** to **22333** once to join

# Now that we have finished the discussion on Slowly Changing Dimensions, I am feeling:

A: Good – The concepts are clear

B: OK – A little review of the slides and I'm in good shape

C: Bad – This doesn't make any sense to me

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**